

Task 5 Data Analysis

Task :

Exploratory Data Analysis (EDA) on Titanic Dataset

Tools Used:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- VS Code / Jupyter Notebook

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Objective:

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to understand its structure, identify missing values, visualize data distributions, analyse relationships among variables, and extract meaningful insights.

Dataset Overview:

```
# Display first 5 rows
df.head()
```

Python

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Description:

The first five rows of the dataset were displayed to understand the available features and verify that the dataset was loaded correctly.

Dataset Information:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype  
---  -
 0   PassengerId     891 non-null    int64  
 1   Survived        891 non-null    int64  
 2   Pclass          891 non-null    int64  
 3   Name            891 non-null    object  
 4   Sex             891 non-null    object  
 5   Age            714 non-null    float64 
 6   SibSp          891 non-null    int64  
 7   Parch          891 non-null    int64  
 8   Ticket         891 non-null    object  
 9   Fare           891 non-null    float64 
10   Cabin          204 non-null    object  
11   Embarked       889 non-null    object  
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Observation:

- Dataset contains 891 rows and 12 columns.

- Numerical and categorical columns are present.
- Missing values exist in Age, Cabin, and Embarked columns.

Statistical Analysis:

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Observation:

The statistical summary provides information such as mean, minimum, maximum, standard deviation, and quartiles for numerical columns.

Missing Values:

```
df.isnull().sum()
```

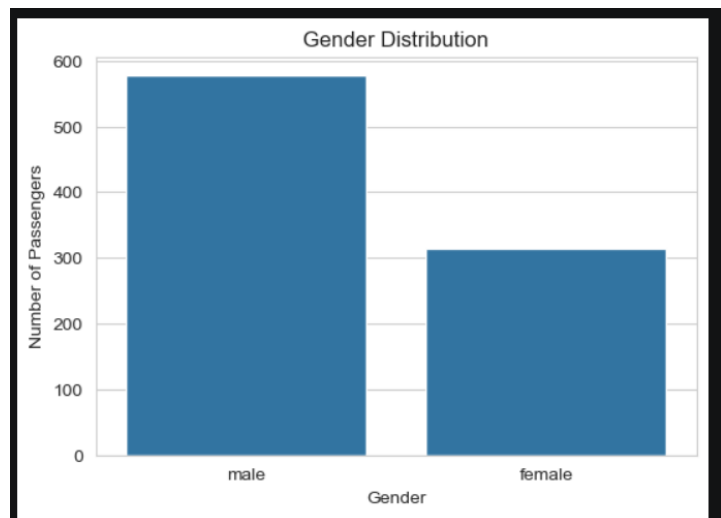
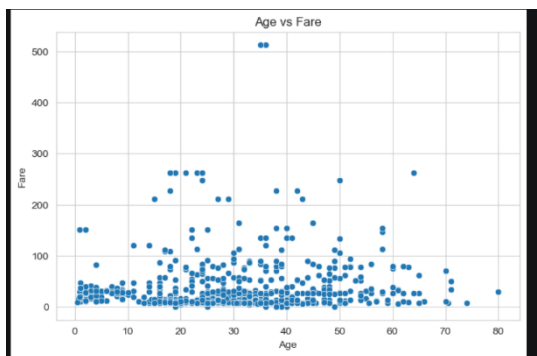
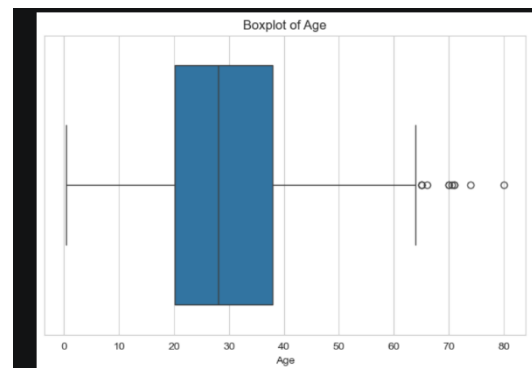
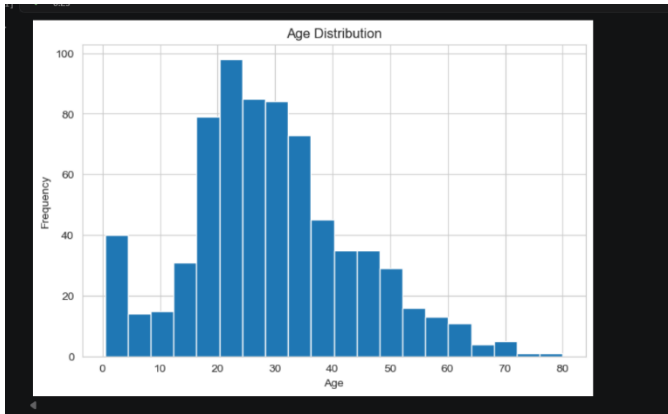
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2
dtype: int64	

Observation:

- Age contains missing values.

- Cabin has the highest number of missing values.
- Embarked contains only a few missing values.

Major Distributions:



Key Findings

- The dataset contains **891 passengers** and **12 features**.
- Missing values were found in Age, Cabin, and Embarked.
- Cabin contains the highest number of missing values.
- Most passengers belonged to Passenger Class 3.
- Male passengers were more than female passengers.
- Most passengers were between 20 and 40 years old.
- More passengers did not survive than survived.
- Fare has a relationship with Passenger Class.
- Age and Fare have no strong correlation.

Conclusion

Exploratory Data Analysis helped identify important patterns, trends, missing values, and relationships within the Titanic dataset. Various statistical methods and visualizations provided meaningful insights that can support future predictive modeling and machine learning tasks.